

Topic 04: Separable Equations & Initial Value Problems (The Clean Split)

Professor Leonard Master Series · Lessons 12 to 14

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Great Divorce

Separable differential equations are the foundation of analytical ODEs. Professor Leonard describes the process as "The Great Divorce":

"You have an equation where x 's and y 's are mingled together. Your mission is to send all y 's to the left side with dy , and all x 's to the right side with dx . Once the divorce is finalized, you integrate both sides independently!"

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)} dy = g(x) dx$$

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 4.1: Polynomial Separation & Cubic Roots

Statement: Solve $\frac{dy}{dx} = \frac{x^2}{y^2}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate:**

$$y^2 dy = x^2 dx$$

- **Step 2: Integrate both sides:**

$$\int y^2 dy = \int x^2 dx \implies \frac{y^3}{3} = \frac{x^3}{3} + C_1$$

- **Step 3: Clear denominators:** Multiply by 3:

$$y^3 = x^3 + 3C_1 \implies y^3 = x^3 + C$$

- **Step 4: Solve explicitly for y :**

$$y(x) = \sqrt[3]{x^3 + C}$$

Problem 4.2: The "Lost" Singular Solution Trap

Statement: Find the complete general solution of $\frac{dy}{dx} = 6x(y-1)^{2/3}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Note potential division by zero:** To separate, we must divide by $(y-1)^{2/3}$. This assumes $y-1 \neq 0$.

Check singular candidate

: If $y(x) \equiv 1$, then $y' = 0$ and $6x(1-1)^{2/3} = 0$. Thus, $y = 1$ is an **exact equilibrium solution!**

- **Step 2: Separate assuming $y \neq 1$:**

$$(y-1)^{-2/3} dy = 6x dx$$

- **Step 3: Integrate both sides:**

$$\frac{(y-1)^{1/3}}{1/3} = 3x^2 + C_1 \implies 3(y-1)^{1/3} = 3x^2 + C_1$$

- **Step 4: Divide by 3 (relabeling $C = C_1/3$):**

$$(y-1)^{1/3} = x^2 + C$$

- **Step 5: Cube both sides:**

$$y-1 = (x^2 + C)^3 \implies y(x) = (x^2 + C)^3 + 1$$

- **Note on Singular Solution:** Can $(x^2 + C)^3 + 1$ produce $y = 1$? Only if $x^2 + C = 0$ for all x , which is impossible for constant C . Therefore, $y = 1$ is a **singular solution** that cannot be obtained from the general family. Full answer: $y(x) = (x^2 + C)^3 + 1$ and $y = 1$.

Problem 4.3: Initial Value Problem with Natural Logarithms

Statement: Solve the IVP:

$$\frac{dy}{dx} = \frac{2x(y^2 + 1)}{y}, \quad y(0) = 1$$

Step-by-Step Whiteboard Solution:

- **Step 1: Separate variables:**

$$\frac{y}{y^2 + 1} dy = 2x dx$$

- **Step 2: Integrate both sides:**

$$\int \frac{y}{y^2 + 1} dy = \int 2x dx$$

LHS: Let $u = y^2 + 1$, $du = 2ydy \implies ydy = \frac{1}{2}du$.

$$\frac{1}{2} \ln(y^2 + 1) = x^2 + C_1$$

(No absolute value needed because $y^2 + 1 > 0$ for all real y)

- **Step 3: Clear the fraction and exponentiate:**

$$\ln(y^2 + 1) = 2x^2 + 2C_1$$

$$y^2 + 1 = e^{2x^2 + 2C_1} = e^{2C_1} \cdot e^{2x^2} = Ae^{2x^2} \quad (A > 0)$$

$$y^2 = Ae^{2x^2} - 1$$

- **Step 4: Apply $y(0) = 1$:**

$$1^2 = Ae^0 - 1 \implies 1 = A - 1 \implies A = 2$$

$$y^2 = 2e^{2x^2} - 1 \implies y(x) = \pm \sqrt{2e^{2x^2} - 1}$$

- **Step 5: Select the correct branch:** Because $y(0) = +1 > 0$, choose the positive square root:

$$y(x) = \sqrt{2e^{2x^2} - 1}$$

Problem 4.4: Trigonometric Separation & Domain Blow-Up

Statement: Solve $\frac{dy}{dx} = y^2 \sin x$, $y(0) = 1$, and find the interval of definition.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate:**

$$y^{-2} dy = \sin x dx$$

- **Step 2: Integrate:**

$$-y^{-1} = -\cos x + C \implies \frac{1}{y} = \cos x - C$$

- **Step 3: Apply $y(0) = 1$:**

$$\frac{1}{1} = \cos(0) - C \implies 1 = 1 - C \implies C = 0$$

- **Step 4: Invert:**

$$\frac{1}{y} = \cos x \implies y(x) = \frac{1}{\cos x} = \sec x$$

- **Step 5: Interval of definition:** The function $\sec x$ blows up to ∞ at $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$. The initial condition is given at $x_0 = 0$. The largest continuous open interval containing 0 is:

$$I = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Problem 4.5: Integration by Substitution on the RHS

Statement: Solve the IVP:

$$\frac{dy}{dx} = \frac{xy^3}{\sqrt{1+x^2}}, \quad y(0) = -1$$

Step-by-Step Whiteboard Solution:

- **Step 1: Separate:**

$$y^{-3}dy = \frac{x}{\sqrt{1+x^2}}dx$$

- **Step 2: Integrate both sides:**

$$\int y^{-3}dy = \int x(1+x^2)^{-1/2}dx$$

LHS: $\frac{y^{-2}}{-2} = -\frac{1}{2y^2}$. RHS: Let $u = 1 + x^2, du = 2xdx$. $\int \frac{1}{2}u^{-1/2}du = u^{1/2} + C = \sqrt{1+x^2} + C$.

$$-\frac{1}{2y^2} = \sqrt{1+x^2} + C$$

- **Step 3: Apply $y(0) = -1$:**

$$-\frac{1}{2(-1)^2} = \sqrt{1+0} + C \implies -\frac{1}{2} = 1 + C \implies C = -\frac{3}{2}$$

- **Step 4: Solve for $y(x)$:**

$$-\frac{1}{2y^2} = \sqrt{1+x^2} - \frac{3}{2} \implies \frac{1}{2y^2} = \frac{3 - 2\sqrt{1+x^2}}{2}$$

$$y^2 = \frac{1}{3 - 2\sqrt{1+x^2}}$$

Since $y(0) = -1 < 0$, choose the **negative square root**:

$$y(x) = -\frac{1}{\sqrt{3 - 2\sqrt{1+x^2}}}$$

Problem 4.6: Implicit Factoring & Integration by Parts

Statement: Solve $e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Factor the RHS:**

$$e^{-y} + e^{-2x-y} = e^{-y}(1 + e^{-2x})$$

- **Step 2: Separate x and y :**

$$e^x y \frac{dy}{dx} = e^{-y}(1 + e^{-2x}) \implies \frac{y}{e^{-y}} dy = \frac{1 + e^{-2x}}{e^x} dx$$

$$ye^y dy = (e^{-x} + e^{-3x}) dx$$

- **Step 3: Integrate LHS by parts:** Let $u = y, dv = e^y dy \implies du = dy, v = e^y$.

$$\int ye^y dy = ye^y - \int e^y dy = (y - 1)e^y$$

- **Step 4: Integrate RHS:**

$$\int (e^{-x} + e^{-3x}) dx = -e^{-x} - \frac{1}{3}e^{-3x} + C$$

- **Step 5: Full implicit general solution:**

$$(y - 1)e^y + e^{-x} + \frac{1}{3}e^{-3x} = C$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Dropping $+C$ inside algebra:** Writing $\ln y = x \implies y = e^x + C$ instead of $y = e^{x+C} = Ae^x$.
- **Trap 2: Ignoring Square Root Signs:** When taking $y = \pm\sqrt{\dots}$, you must check the initial condition $y(x_0) = y_0$ to choose the correct $+/ -$ sign.
- **Trap 3: Forgetting Singular Solutions:** Always check if dividing by $h(y)$ throws away an equilibrium solution $h(y) = 0$.